# Project 1 — One Problem, Three Lenses (Modules 1–4)

## INFO 521 — Machine Learning Foundations

> ℹ Course-schedule placement
>
> **Assigned Week 1 · due Week 4.** Milestones on the module arc: 1.1 Least Squares (M1–M2, Wks 1–2) → 1.2 Maximum Likelihood (M3, Wk 3) → 1.3 Bayesian (M4, Wk 4 — the conjugate Gaussian posterior is the Project-1 endpoint) → 1.4 Synthesis (Wk 4). Gating: Checkpoints 1–3.

One clinical dataset, one prediction problem (age → systolic blood pressure), examined through three accreting lenses and then synthesized.

## Milestones

1. **1.1 Least Squares** (M1–M2) — design matrix, normal equations, K-fold CV, L2. Play artifact: a 3×3 bias–variance sweep over polynomial orders {1, 3, 9} and λ ∈ {1e-3, 1e-1, 1e1}. Gated by Checkpoint 1.
2. **1.2 Maximum Likelihood** (M3) — Gaussian-noise generative model, MLE for **w** (shown numerically equal to the 1.1 OLS fit), $\hat{\sigma}^2$, predictive error bars. Play artifact: where the model is least sure, and why. Gated by Checkpoint 2.
3. **1.3 Bayesian** (M4) — Gaussian prior, conjugate posterior, posterior-predictive credible intervals. Play artifact: prior-strength and data-amount sweeps showing collapse toward the MLE. Gated by Checkpoint 3.
4. **1.4 Synthesis** — one panel, three lenses; argued compare-and-contrast; reproducible end-to-end. Structured peer review.

## Through-line

Project 1 stays **single-predictor polynomial** the whole way (model order is the only complexity knob, exactly the Module 1–2 lecture arc). Expanding to the full multivariate feature set is one of the things **Project 2 adds**.

## Satisfactory

Each notebook states its own criteria at the top. The project is complete when 1.1–1.4 are all Satisfactory and Checkpoints 1–3 are passed. See `../GRADING.md`.

## Bridge to Project 2

1.3 leans entirely on conjugacy (closed-form Gaussian posterior). Project 2 opens by binarizing systolic BP into a **hypertension** label (ACC/AHA: SBP ≥ 130 or DBP ≥ 80) — conjugacy breaks, the posterior goes intractable, and approximate inference becomes necessary. The same binary label, predicted from the **non-BP** features (the blood-pressure columns are held out to avoid label leakage), then drives classification; dropping it entirely motivates clustering and PCA.